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Latch structure and optical receptacle thereof

Abstract

The present invention provides a latch structure arranged in an optical receptacle. The latch structure comprises a supporting element and a first clip structure, wherein the supporting element is assembled with the housing of the optical receptacle. The first clip structure is formed on the supporting element for retaining the optical connector inserting into the optical receptacle. The main idea of the present invention is that when the latch structure assembling with the optical receptacle, there is no interaction force between the first clip structure and optical receptacle so as to maintain the clipping force of the first clip structure acting on the optical connector.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit under 35 U.S.C. § 119 (e) to the U.S. provisional patent application having the Ser. No. 62/981,537 filed on Feb. 26, 2020, the entirety of which is incorporated herein by reference.

FIELD OF INVENTION
(1) The present invention relates to a latch structure. In particular, it relates to a latch structure used for coupling to an optical connector and an optical receptacle thereof.
BACKGROUND OF THE INVENTION
(2) Due to the advantages of high frequency bandwidth and low loss, optical fibers have been widely used as signal transmission media in recent years. The use of optical fiber has already had a major revolutionary impact in the communications industry. Nowadays, 100G optical module communication is not enough, and the future will be expected to move towards the era of 400G optical module communications.
(3) In the field of 400G optical communications, there are also many designs for the packaging design of optical fiber modules, one of which is called Quad Small Form Factor Pluggable-Double Density (QSFF-DD). The specification, with a downward compatible design, has attracted the attention of many large manufacturers, and has launched corresponding specifications of optical communication module products.
(4) The above information disclosed in this section is only for enhancement of understanding of the background of the described technology and therefore it may contain information that does not form the prior art that is already known to a person of ordinary skill in the art.
SUMMARY OF THE INVENTION
(5) The present invention provides a latch structure and an optical receptacle thereof. When latch structure is assembled inside the housing of the optical receptacle, the housing of the optical receptacle and the latch structure are used to clip the optical connector inserted from the opening of the optical receptacle. There is no force generated between the clip structures, so that the pair of clip elements on the clip structure can maintain the same distance, thereby maintaining the clipping

effect.
(6) In one embodiment of the present invention, the present invention provides a latch structure, disposed in an optical receptacle, comprising: a supporting element, configured for combining with the optical receptacle; and a first clip structure, formed on the supporting element, and configured for buckling with an optical connector, wherein during combination process of the supporting element and the optical receptacle, there is no interaction force between the optical receptacle and the first clip structure.
(7) In another embodiment of the present invention, the present invention also provides an optical receptacle, comprising: a housing, having an accommodation space, a first assembly structure disposed in the accommodation space of the housing; and a latch structure, disposed in the accommodation space, comprising: a supporting element, configured for combining with the first assembly structure; and a first clip structure, formed on the supporting element, and configured for buckling with an optical connector, wherein during combination process of the supporting element and the first assembly structure, there is no interaction force between the housing and the first clip structure.
(8) Many of the attendant features and advantages of the present invention will become better understood with reference to the following detailed description considered in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The detailed structure, operating principle and effects of the present invention will now be described in more details hereinafter with reference to the accompanying drawings that show various embodiments of the present invention as follows.
(2) FIG. **1** is a three-dimensional schematic diagram of an embodiment of the latch structure in the present invention.
(3) FIG. **2** is a three-dimensional cross-sectional schematic diagram of another embodiment of the latch structure in the present invention.
(4) FIG. **3** is a three-dimensional schematic diagram of other embodiment of the latch structure in the present invention.
(5) FIG. **4**A is a three-dimensional and exploded schematic diagram of an embodiment of the optical receptacle in the present invention.
(6) FIG. **4**B is a schematic cross-sectional diagram of an embodiment of the optical receptacle in the present invention.
(7) FIG. **5**A is a three-dimensional assembly diagram of an embodiment of the optical receptacle in the present invention.
(8) FIG. **5**B is a schematic cross-sectional diagram of the combination of the housing and the latch structure of an embodiment of the optical receptacle in the present invention.
(9) FIG. **6**A is a three-dimensional schematic diagram of another embodiment of the optical receptacle in the present invention.
(10) FIG. **6**B is a schematic cross-sectional diagram of another embodiment of the optical receptacle in the present invention.
(11) FIG. **6**C is an exploded schematic diagram of the housing section and the latch structure of another embodiment of the optical receptacle in the present invention.
(12) FIG. **7**A is a three-dimensional and exploded schematic diagram of another embodiment of the optical receptacle and optical connector in the present invention.
(13) FIG. **7**B is a schematic cross-sectional diagram of another embodiment of the optical receptacle shown in FIG. **7**A.

(14) FIG. **7**C is an exploded schematic diagram of the housing section and the latch structure of another embodiment of the optical receptacle in the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(15) Reference will now be made in detail to the exemplary embodiments of the present invention, examples of which are illustrated in the accompanying drawings. Therefore, it is to be understood that the foregoing is illustrative of exemplary embodiments and is not to be construed as limited to the specific embodiments disclosed, and that modifications to the disclosed exemplary embodiments, as well as other exemplary embodiments, are intended to be included within the scope of the appended claims. These embodiments are provided so that this invention will be thorough and complete, and will fully convey the inventive concept to those skilled in the art. The relative proportions and ratios of elements in the drawings may be exaggerated or diminished in size for the sake of clarity and convenience in the drawings, and such arbitrary proportions are only illustrative and not limiting in any way.

(16) For convenience, certain terms employed in the specification, examples and appended claims are collected here. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of the ordinary skill in the art to which this invention belongs.

(17) Various embodiments will now be described more fully with reference to the accompanying drawings, in which illustrative embodiments are shown. The inventive concept, however, may be embodied in various different forms, and should not be construed as being limited only to the illustrated embodiments. Rather, these embodiments are provided as examples, to convey the inventive concept to one skilled in the art. Accordingly, known processes, elements, and techniques are not described with respect to some of the embodiments.

(18) The singular forms "a", "and", and "the" are used herein to include plural referents unless the context clearly dictates otherwise.

(19) The following descriptions are provided to elucidate a latch structure and optical receptacle thereof and to aid it of skilled in the art in practicing this invention. These embodiments are merely exemplary embodiments and in no way to be considered to limit the scope of the invention in any manner.

(20) Please refer to FIG. **1**, which is a three-dimensional schematic diagram of an embodiment of the latch structure in the present invention. In this embodiment, the latch structure **2** includes a supporting element **20**, a first clip structure **21** and a second assembly structure **22**. In this embodiment, the supporting element **20**, the first clip structure **21**, and the second assembly structure **22** are integrally formed, but are not limited thereto. For example: the supporting element **20**, the first clip structure **21**, and the second assembly structure **22** may also be an independent element, which make up the latch structure **2**. The first clip structure **21** further has a first fastener **210**A and a second fastener **210**B, which are respectively located at the two end portions E**1** and E**2** of the supporting element **20** and extended toward the first side **20**A. In this embodiment, the supporting element **20** has a first through hole **200** and a second through hole **201** on the surface on the second side **20**B. The second assembly structure **22** has a first assembly element **220**A and a second assembly element **220**B, which are respectively disposed on the two end portions E**1** and E**2** of the supporting element **20**. In this embodiment, the first assembly element **220**A and the second assembly element **220**B are respectively groove structures. The first assembly element **220**A is disposed on one side of the first through hole **200** and corresponding to the first fastener **210**A, and the second coupling element **220**B is disposed on one side of the second through hole **201** and corresponding to the second fastener **210**B. In this embodiment, the first side **20**A of the supporting element **20** further has a first coupling structure **24** formed between the first fastener **210**A and the second fastener **210**B. The first coupling structure **24** has a coupling hole communicating with the first through hole **200** and the second through hole **201**.

(21) Please refer to FIG. **2**, which is a three-dimensional cross-sectional schematic diagram of

another embodiment of the latch structure in the present invention. In this embodiment, the latch structure **3** includes a supporting element **30**, a first clip structure **31** and a second assembly structure **32**. In this embodiment, the supporting element **30**, the first clip structure **31**, and the second assembly structure **32** are integrally formed, but not limited thereto. For example: the supporting element **30**, the first clip structure **31**, and the second assembly structure **32** may also be an independent element, which make up the latch structure **3**. In this embodiment, the first clip structure **31** includes a first fastener **310**A and a second fastener **310**B, which are respectively located at the two end portions E**1** and E**2** of the supporting element **30** and extended toward the first side **30**A. The second assembly structure **32** is disposed on the supporting element **30**. In this embodiment, the second assembly structure **32** is disposed on the end portion E**1** of the supporting element **30** and corresponds to the first fastener **310**A, and the second assembly structure **32** is a groove structure. In this embodiment, the supporting element **30** has a first through hole **300** and a second through hole **301** on the surface on the second side **30**B. The second assembly structure **32** is disposed on one side of the first through hole **300**. At a position where the surface of the second side **30**B of the supporting element **30** is close to the second fastener **310**B, there is a third assembly structure **33** extended from the surface of the end portion E**2** of the supporting element **30** toward the second side **30**B. A hook portion **330** is disposed on the end of the third assembly structure **33** to generate the fixing and positioning effect. In this embodiment, the first side **30**A of the supporting element **30** further has a first coupling structure **34** formed between the first fastener **310**A and the second fastener **310**B. The first coupling structure **34** has a coupling hole communicating with the first through hole **300** and the second through hole **301**.

(22) Please refer to FIG. **3**, which is a three-dimensional schematic diagram of other embodiment of the latch structure in the present invention. In this embodiment, the latch structure **4** includes a supporting element **40**, a first clip structure **41** and a second assembly structure **42**. In this embodiment, the supporting element **40**, the first clip structure **41**, and the second assembly structure **42** are integrally formed, but it is not limited thereto. For example, in another embodiment, the supporting element **40**, the first clip structure **41** and the second assembly structure **42** may also be independent elements, which make up the latch structure **4**. The first clip structure **41** is formed on the surface of the supporting element **40** on the first side **40**A. In this embodiment, the first clip structure **41** further includes a first fastener **410**A and a second fastener **410**B respectively located at the two end portions E**1** and E**2** of the supporting element **40** and extended toward the first side **40**A. The second assembly structure **42** is disposed on the surface of the supporting element **40** on the second side **40**B. In this embodiment, the second assembly structure **42** extends from the surface of the supporting element **40** toward the second side **40**B.

(23) In this embodiment, the second assembly structure **42** is disposed on one side of the first through hole **400**. There is a third coupling structure **43** at the end portion E**2** of the supporting element **40** extended from the surface of the supporting element **40** toward the second side **40**B. A hook portion **430** is disposed on the end of the third assembly structure **43** to generate a fixing and positioning effect. In this embodiment, the supporting element **40** on the first side **40**A further has a first coupling structure **44** formed between the first fastener **410**A and the second fastener **410**B. The first coupling structure **44** has a coupling hole communicating with the first through hole **400** and the second through hole **401**. It should be noted that the type of the second assembly structure or the third assembly structure in the foregoing embodiments is not limited, and the main concept is to combine the latch structures **2** to **4** in the optical receptacle. In addition, during the assembling process, the first fasteners **210**A, **310**A, and **410**A and the corresponding second fasteners **210**B, **310**B, and **410**B would not be suffered from any force, and the distance between those two would not be changed.

(24) Please refer to FIGS. **4**A and **4**B, in which FIG. **4**A is a three-dimensional and exploded schematic diagram of an embodiment of the optical receptacle in the present invention, and FIG. **4**B is a schematic cross-sectional diagram of an embodiment of the optical receptacle in the present

invention. The optical receptacle **5** of the present invention is used for accommodating the optical connectors C**1** and C**2** from the first opening **5**A and the second opening **5**B on both sides. The optical receptacle **5** includes a housing **50** and a latch structure **2**. The housing **50** has an accommodation space **5**C. There are first assembly structures **51**A and **51**B on one of the side walls of the housing **50** in the accommodation space **5**C. In this embodiment, the first assembly structures **51**A and **51**B respectively form the upper side and the lower side of the housing **50**, and each first assembly structure **51**A and **51**B is a flexible cantilever structure with an engaging structure **510**A and **510**B at those free end. The optical receptacle **5** has a second coupling structure **53** for coupling with the optical connector C**1**. The second coupling structure **53** is a hollow cylindrical structure and is formed on the side of the first opening **5**A of the housing **50**.

(25) In this embodiment, the latch structure **2** is the same as that in FIG. **1**. In this embodiment, the latch structure **2** is disposed in the accommodation space **5**C from the second opening **5**B of the optical receptacle **5**, and is buckled with the first coupling structures **51**A and **51**B. In this embodiment, the first assembly element **220**A of the second assembly structure **22** on the supporting element **20** of the latch structure **2** is coupled with the first assembly structure **51**A in the optical receptacle **5**. The second coupling element **220**B disposed at another end of the supporting element **20** is coupled with the first assembly structure **51**B in the optical receptacle **5**. The first assembly element **220**A and the second assembly element **220**B of this embodiment are grooved structures, and the engaging structures **510**A and **510**B at the ends of the first assembly structures **51**A and **51**B are hook structures, so that the engaging structures **510**A and **510**B can be snapped into the grooves of the second assembly elements **220**A and **220**B. In addition, the first coupling structure **24** between the first fastener **210**A and the second fastener **210**B of the latch structure **2** is used to electrically connect to the optical connector C**2** inserted into the optical receptacle **5** from the second opening **5**B. The first coupling structure **24** has through holes **240** and **241** for allowing the optical fiber terminal at the end of the optical connector C**2** to pass through. It is related to the prior art and would not be repeated herein.

(26) Next, the main concept of FIGS. **1** and **4**A and **4**B in the present invention is going be explained. In the embodiment of the present invention, when the latch structure **2** is installed in the optical receptacle **5**, the first fastener **210**A and the second fastener **210**B of the first clip structure **21** is not suffered from the interaction force of the housing of the optical receptacle **5**, and the buckle method is through the combination of the first assembly structure **51**A and the first assembly element **220**A and the combination of the first assembly structure **51**B and the second assembly element **220**B, it enables the latch structure **2** to be firmly fixed into the optical connector **5**. When the latch structure **2** is installed in the optical receptacle **5**, this embodiment may prevent the first fastener **210**A and the second fastener **210**B of the first clip structure **21** from squeezing by the internal housing of the optical receptacle **5**, which causes stress or deformation, so that the distance between the first fastener **210**A and the second fastener **210**B is not changed, thereby not affecting the clipping and fixing effect of the first fastener **210**A and the second fastener **210**B for clipping the inserted optical connector C**2** from the second opening **5**B.

(27) Please refer to FIGS. **5**A and **5**B, in which FIG. **5**A is a three-dimensional assembly diagram of an embodiment of the optical receptacle in the present invention, and FIG. **5**B is a schematic cross-sectional diagram of the combination of the housing and the latch structure of an embodiment of the optical receptacle in the present invention. The optical receptacle **6** of the present invention is used to allow optical connectors (not shown) to be inserted from the first opening **6**A and the second opening **6**B on both sides, and the second coupling structure **62** in the optical receptacle **6** and the first coupling structure **34** of the latch structure **3** enable the optical connectors inserted from the first opening **6**A and the second opening **6**B to be electrically coupled to each other.

(28) The optical receptacle **6** includes a housing **60** and a latch structure **3** installed in the housing **60**. The side wall in the housing **60** has a first assembly structure **61**A and a fourth assembly structure **61**B that are combined with the latch structure **3**. In this embodiment, the first assembly

structure **61**A is a cantilever structure with a snap **610** at its end. The fourth assembly structure **61**B is a convex structure. It should be noted that the types of the first assembly structure **61**A and the fourth assembly structure **61**B are not limited thereto. For example: in another embodiment, the first assembly structure **61**A may be a convex structure, and the fourth assembly structure **61**B may be a cantilever structure. Alternatively, the first assembly structure **61**A may also be a cantilever structure, and the fourth assembly structure **61**B is a trench structure, etc., which may be changed by those skilled in the art.

(29) In this embodiment, the second coupling structure **62** in the optical receptacle **6** is used for coupling with the optical connector. The second coupling structure **62** has a first hollow cylindrical structure **62**A and a second hollow cylindrical structure **62**B, which are used to couple with the optical fiber terminal of the optical connector. In this embodiment, the housing **60** has a plurality of slots **600˜602**, and each slot **600˜602** corresponds to the second coupling structure **62**, and each slot **600˜602** may allow the optical connector for insertion. The number of slots is determined according to needs, and is not limited herein.

(30) In addition, there is at least one first positioning structure **606** on the outer surface of the first opening **6**A of the housing **60**. The outer surface of the housing **60** is covered with a first cover housing **64** which has at least one second positioning structure **643** corresponding to the first positioning structure **606** respectively. When the first cover housing **64** is combined with the outer surface of the first opening **6**A of the housing **60**, each second positioning structure **643** is combined with the corresponding first positioning structure **606**. In this embodiment, the first cover housing **64** is a metal structure, and a notch **640** is disposed on one side wall of the first cover housing **64**, so that the side walls form an upper lateral plate **641** and a lower lateral plate **642**, and the upper lateral plate **641** and lower lateral plate **642** respectively have a second positioning structure **643**. In this embodiment, the second positioning structure **643** is an opening structure, and the first positioning structure **606** is a corresponding convex structure that may be embedded in the opening structure. The position that the housing **60** corresponding to the notch **640** is an assembly block **604** for combining with the notch **640** so that the first cover housing **64** may be fixed on the housing **60**. In addition, there are a plurality of convex plate body **605** on one side of the first opening **6**A, and the first cover housing **64** has a covering member **644**, which corresponds to each convex plate body **605** and is connected to the end of the convex plate body **605**.

(31) At least one third positioning structure **603** is disposed on the outer surface of the housing **60** at the second opening **6**B. The second cover housing **65** has at least one fourth positioning structure **650** corresponding to the third positioning structure **603** respectively. When the second cover housing **65** is combined with the outer surface of the housing **60** at the second opening **6**B, each fourth positioning structure **650** is combined with the corresponding third positioning structure **603**. In this embodiment, the second cover housing **65** is a metal structure, and a fourth positioning structure **650** is formed on both side walls thereof. In this embodiment, the fourth positioning structure **650** is an opening structure, and the third positioning structure **603** is a corresponding convex structure that may be embedded in the opening structure. In addition, a first stopper **651** is formed on one side of the fourth positioning structure **650** in this embodiment, and a second stopper **652** is formed on an open end of the second cover housing **65**. The usage of the first stopper **651** and the second stopper **652** is that when the entire optical receptacle **6** is inserted into a shell of an apparatus, the first stopper **651** and the second stopper **652** can be positioned on the front and rear sides of the shell, and generate the fixing effect.

(32) Next, the method of installing the latch structure **3** into the housing **60** is going to be explained. In this embodiment, the second assembly structure **32** on one end of the supporting element **30** is a groove structure. Therefore, when the latch structure **3** is inserted into the housing **60** through the first opening **6**A, it may be buckled with the snap **610** of the first assembly structure **61**A in the housing **60**. The third assembly structure **33** at another end of the supporting element **30** is coupled with the fourth assembly structure **61**B in the housing **60**. One end of the third assembly

structure **33** in this embodiment is a hook portion **330**, when the latch structure **3** is installed, the hook portion **330** and the fourth assembly structure **61**B would be clipped together. By the combination of the first assembly structure **61**A and the second assembly structure **32** and the combination of the third assembly structure **33** and the fourth assembly structure **61**B, the latch structure **3** may be firmly fixed in the housing **60**. After the latch structure **3** is assembled into the housing **60**, the first coupling structure **34** on the latch structure **3** corresponds to the second coupling structure **62** in the second opening **6**B of the housing **60**. The first through hole **300** and the second through hole **301** on one side of the first coupling structure **34** correspond to the through holes in the first hollow cylindrical structure **62**A and the second hollow cylindrical structure **62**B of the second coupling structure **62**. Therefore, the optical connectors inserted from the first opening **6**A and the second opening **6**B may be coupled to each other by the first coupling structure **34** and second coupling structure **62**.
(33) The latch structure **3** is inserted into the housing **60** from the first opening **6**A by the combination of the first assembly structure **61**A and the second assembly structure **32** on the support element **30**, and by the combination of the fourth assembly structure **61**B and the third assembly structure **33** on the supporting element **30**. Therefore, when the latch structure **3** is installed into the optical receptacle **6**, the first fastener **310**A and the second fastener **310**B of the first clip structure **31** would not be suffered from the interaction force of the housing of the optical receptacle **6**. In this way, not only the latch structure **3** may be firmly assembled in the housing **60**, but also when the latch structure **3** is installed into the optical receptacle **6**, the first fastener **310**A and the second fastener **310**B of the first clip structure **31** is prevented from squeezing by the housing of the optical receptacle **6**, which causes stress or deformation, so that the distance between the first fastener **310**A and the second fastener **310**B is not changed, thereby not affecting the clipping and fixing effect of the first fastener **310**A and the second fastener **310**B for clipping the inserted optical connector from the opening **6**A.
(34) Please refer to FIGS. **6**A to **6**C, in which FIG. **6**A is a three-dimensional schematic diagram of another embodiment of the optical receptacle in the present invention, FIG. **6**B is a schematic cross-sectional diagram of another embodiment of the optical receptacle in the present invention, and FIG. **6**C is an exploded schematic diagram of the housing section and the latch structure of another embodiment of the optical receptacle in the present invention. The optical receptacle **7** of this embodiment has a housing **7**A and a latch structure **4** as shown in FIG. **3**. The housing **7**A has an accommodation space **7**B, and a first assembly structure **70** is disposed on one of the side walls of the housing **7**A in the accommodation space **7**B. In this embodiment, the first assembly structure **70** has a supporting plate **700** and an assembly element **701**A. The supporting plate **700** has a first notch **700**A communicating with the groove inside the assembly element **701**A.
(35) Through the combination of the second assembly structure **42** and the first assembly structure **70**, and the combination of the hook portion **430** of the third assembly structure **43** and the fourth assembly structure **72** of the optical receptacle **7**, the latch structure **4** may be firmly disposed in the optical receptacle **7**. In this embodiment, the second assembly structure **42** passes through the first notch **700**A and then is combined with the assembly element **701**A to form a hollow cylindrical structure as the second coupling structure **71** coupled to the optical connector. After the latch structure **4** is combined with the first assembly structure **70**, the supporting plate **700** is leaned against the supporting element **40**.
(36) Next, the main concept of FIGS. **6**A to **6**C is going be explained. In the embodiment of the present invention, when the latch structure **4** is installed in the optical receptacle **7**, the first fastener **410**A and the second fastener **410**B of the first clip structure **41** is not suffered from the interaction force of the housing of the optical receptacle **7**, and the buckle method is through the interaction force between the first assembly structure **70** and the second assembly structure **42**, and the interaction force between the hook portion **430** and the fourth assembly structure **72** in the optical receptacle **7**. It enables the latch structure **4** to be firmly arranged in the housing **7**A. Through this

way, not only the latch structure **4** may be firmly assembled in the housing **7**A, but also when the latch structure **4** is installed into the optical receptacle **7**, the first fastener **410**A and the second fastener **410**B of the first clip structure **41** is prevented from squeezing by the housing of the optical receptacle **7**, which causes stress or deformation, so that the distance between the first fastener **410**A and the second fastener **410**B is not changed, thereby not affecting the clipping and fixing effect of the first fastener **410**A and the second fastener **410**B for clipping the inserted optical connector.

(37) Please refer to FIGS. **7**A to **7**C, in which FIG. **7**A is a three-dimensional and exploded schematic diagram of another embodiment of the optical receptacle and optical connector in the present invention, FIG. **7**B is a schematic cross-sectional diagram of another embodiment of the optical receptacle shown in FIG. **7**A, and FIG. **7**C is an exploded schematic diagram of the housing section and the latch structure of another embodiment of the optical receptacle in the present invention. The optical receptacle **9** of this embodiment has a housing **9**A and a latch structure **8**. The housing **9**A has an accommodation space **9**B, and a first assembly structure **90** is disposed on one of the side walls of the housing **9**A in the accommodation space **9**B. In this embodiment, the first assembly structure **90** has a supporting plate **900** and assembly elements **901**A and **901**B. The supporting plate **900** has notches **900**A and **900**B, which respectively communicate with the grooves inside the assembly elements **901**A and **901**B.

(38) Through the combination of the second assembly structure **82** and the first assembly structure **90**, the latch structure **8** may be firmly disposed in the housing **9**A. In this embodiment, the second assembly structure **82** has a first assembly element **82**A and a second assembly element **82**B, which respectively pass through the notches **900**A and **900**B, and then are combined with the assembly elements **901**A and **901**B to form a hollow cylindrical structure as the second coupling structure **91** coupled to the optical connector. After the latch structure **8** is combined with the first assembly structure **90**, the supporting plate **900** is leaned against the supporting element **80**. In this embodiment, the housing **9**A is installed in the opening hole **920** on the fixing plate **92**. The latch structure **8** further has a first coupling structure **83**, which is formed on another side of the supporting element **80** between the first fastener **81**A and the second fastener **81**B of the first clip structure **81**, that is, on the side opposite to the second assembly structure **82**, and the first coupling structure **83** is used for coupling with a coupling head **93** of the optical connector **90**.

(39) Next, the main concept of FIGS. **7**A to **7**C is going be explained. In the embodiment of the present invention, when the latch structure **8** is installed in housing **9**A of the optical receptacle **9**, the first fastener **81**A and the second fastener **81**B of the first clip structure **81** is not suffered from the interaction force from the inner wall or the inner structure of the housing **9**A. When the latch structure **8** is installed into the housing **9**A, by the interaction force between the first assembly structure **90** and the second assembly structure **82**, the latch structure **8** is firmly combined with the first assembly structure **90**. During the assembly process, the first fastener **81**A and the second fastener **81**B of the first clip structure **81** is prevented from squeezing by the housing **9**A, which causes stress or deformation, so that the distance between the first fastener **81**A and the second fastener **81**B is not changed, thereby maintaining the clipping and fixing effect of the first fastener **81**A and the second fastener **81**B for clipping the inserted optical connector.

(40) It will be understood that the above description of embodiments is given by way of example only and that various modifications may be made by those with ordinary skill in the art. The above specification, examples, and data provide a complete description of the present invention and use of exemplary embodiments of the invention. Although various embodiments of the invention have been described above with a certain degree of particularity, or with reference to one or more individual embodiments, those with ordinary skill in the art could make numerous alterations or modifications to the disclosed embodiments without departing from the spirit or scope of this invention.

Claims

1. A latch structure, disposed in an optical receptacle, comprising: a supporting element, having a plurality of through holes, a first side, and a second side opposite to the first side, and configured for combining with the optical receptacle; a first clip structure, formed on the supporting element and extended toward the first side, and configured for buckling with an optical connector; a second assembly structure, disposed on the supporting element and extended toward the second side, separated by a first predetermined distance from an end of the supporting element, and disposed between the plurality of through holes and the end of the supporting element; and a third assembly structure, disposed on the supporting element and extended toward the second side, separated by a second predetermined distance from another end of the supporting element, and disposed between the plurality of through holes and the another end of the supporting element.
2. The latch structure of claim 1, wherein a first coupling structure is disposed between the first clip structures for electrically connecting with the optical connector.
3. The latch structure of claim 1, wherein the first clip structure has a first fastener and a second fastener, and a first coupling structure is disposed between the first fastener and the second fastener for electrically connecting with the optical connector.
4. An optical receptacle, comprising: a housing, having an accommodation space, a first assembly structure disposed into the accommodation space of the housing; and a latch structure, disposed in the accommodation space, comprising: a supporting element, having a plurality of through holes, a first side, and a second side opposite to the first side, and combining with the optical receptacle; a first clip structure, formed on the supporting element and extended toward the first side, and configured for buckling with an optical connector; a second assembly structure, disposed on the supporting element and extended toward the second side, separated by a first predetermined distance from an end of the supporting element, and disposed between the plurality of through holes and the end of the supporting element; and a third assembly structure, disposed on the supporting element and extended toward the second side, separated by a second predetermined distance from another end of the supporting element, and disposed between the plurality of through holes and the another end of the supporting element.
5. The optical receptacle of claim 4, wherein a first coupling structure is disposed between the first clip structures for electrically connecting with the optical connector.
6. The optical receptacle of claim 4, wherein a first coupling structure is disposed between the first clip structures, and a second coupling structure corresponding to the first coupling structure is disposed in the housing.
7. The optical receptacle of claim 4, comprising a first cover housing which is combined with a first side of the housing, a lateral plate of the first cover housing has a notch for dividing the lateral plate into a first sub-lateral plate and a second sub-lateral plate, which are not connected.
8. The optical receptacle of claim 7, wherein at least one first positioning structure is disposed on a side of a first opening of the housing, the first cover housing has at least one second positioning structure corresponding to the at least one first positioning structure, and the each second positioning structure is combined with the corresponding first positioning structure when the first cover housing is combined with the housing.
9. The optical receptacle of claim 4, comprising a second cover housing, which is combined with a second side of the housing.
10. The optical receptacle of claim 4, wherein the first clip structure has a first fastener and a second fastener, and a first coupling structure is disposed between the first fastener and the second fastener for electrically connecting with the optical connector.